HRMS

12 通知服務 — 系統設計書

HR12 NOTIFICATION Service — System Design

系統名稱	版本
人力資源暨專案管理系統	**1.0**
日期	作者
2026-03-18	**HRMS 開發團隊**

目錄

通知服務系統設計書

版本: 1.0 **日期:** 2025-12-07 **Domain代號:** 12 (NTF) **導入階段:** 第一階段(核心基礎服務) **目標:** 提供工程師完整的系統實作規格,供PM建立工項清單

目錄

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1. 服務概述

1.1 核心功能

- ✅ **多渠道通知:** Email、系統訊息、推播、Teams/LINE
- ✅ **通知範本:** 變數替換、多語系
- ✅ **事件訂閱:** 訂閱所有業務事件發送通知
- ✅ **自動提醒Job:** 生日、合約到期、證照到期
- ✅ **通知偏好:** 使用者個人設定

1.2 通知渠道

渠道	技術	說明
IN_APP	WebSocket	即時系統內通知
EMAIL	Spring Mail + SMTP	HTML郵件
PUSH	Firebase FCM	iOS/Android推播
TEAMS	Webhook	Microsoft Teams整合
LINE	LINE Notify API	LINE整合

2. UI設計

頁面代碼	頁面名稱	路由
`HR12-P01`	通知範本管理頁面	`/admin/notifications/templates`
`HR12-P02`	我的通知頁面	`/profile/notifications`
`HR12-P03`	通知偏好設定頁面	`/profile/notification-settings`

2.1 UI線稿

我的通知頁面 (HR12-P02)

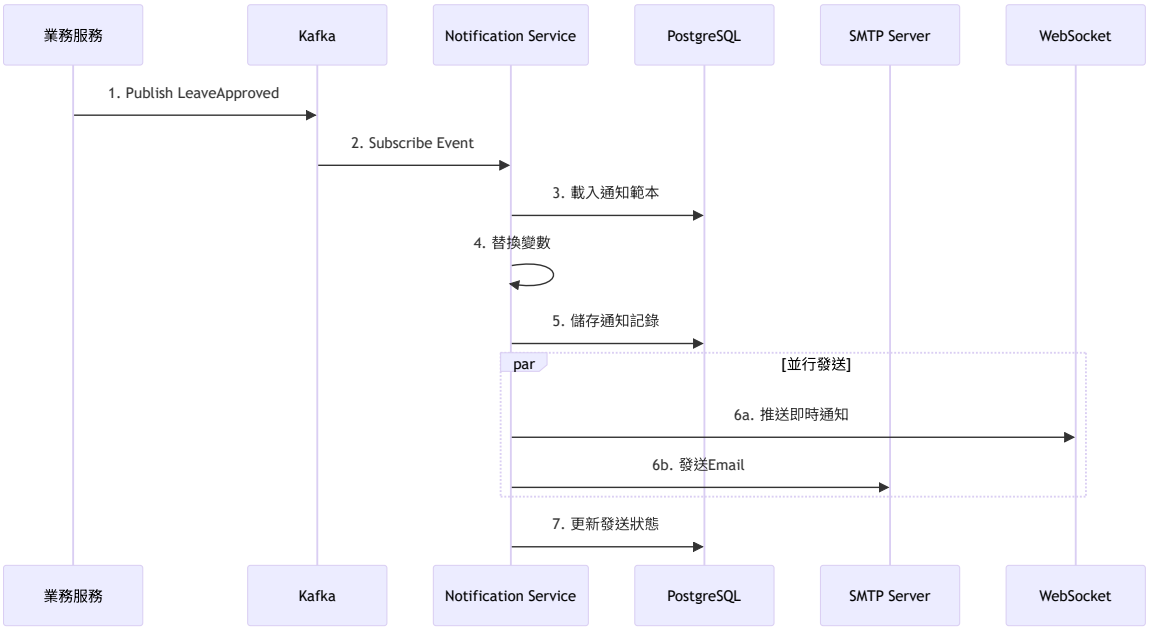
```
┌──────────────────────────────────────────────────────────────┐
│                          我的通知頁面                          │
│                  ┌──────────────────────────────┐              │
│                  │ 我的通知 🔔 3則未讀            │              │
│                  │ ──────────────────────────────│              │
│                  │                               │              │
│                  └──────────────────────────────┘              │
│                                  │                             │
│                                  ▼                             │
│                  ┌──────────────────────────────┐              │
│                  │ [全部] [未讀] [已讀]           │              │
│                  └──────────────────────────────┘              │
│                                  │                             │
│                                  ▼                             │
│  ┌──────────────────────────────────────────────────────────┐  │
│  │                        通知列表                           │  │
│  │                                                           │  │
│  │ ┌─────────────────────────────────────────────────────┐   │  │
│  │ │ ● 【審核請求】員工張三申請特休2天，請審核              │   │  │
│  │ │   12/06 14:30 | [查看詳情]                            │   │  │
│  │ ├─────────────────────────────────────────────────────┤   │  │
│  │ │ ● 【薪資通知】您的2025年11月薪資單已產生               │   │  │
│  │ │   12/05 09:00 | [查看詳情]                            │   │  │
│  │ ├─────────────────────────────────────────────────────┤   │  │
│  │ │ ○ 【審核結果】您的請假申請已核准                       │   │  │
│  │ │   12/04 16:00 | 已讀                                  │   │  │
│  │ └─────────────────────────────────────────────────────┘   │  │
│  │                                                           │  │
│  │                    ● 未讀  ○ 已讀                         │  │
│  └──────────────────────────────────────────────────────────┘  │
└──────────────────────────────────────────────────────────────┘
```

3. Event-Driven架構 (事件驅動)

3.1 事件訂閱對照表

業務事件	通知對象	範本代碼	渠道
`LeaveApplied`	直屬主管	LEAVE_APPROVAL_REQUEST	IN_APP, EMAIL
`LeaveApproved`	申請人	LEAVE_APPROVED	IN_APP, EMAIL
`LeaveRejected`	申請人	LEAVE_REJECTED	IN_APP, EMAIL
`PayslipGenerated`	員工	PAYSLIP_READY	IN_APP, EMAIL
`TaskAssigned`	審核人	APPROVAL_REQUEST	IN_APP, EMAIL
`CertificateExpiring`	員工	CERTIFICATE_EXPIRY	IN_APP, EMAIL
`ContractExpiring`	HR	CONTRACT_EXPIRY	IN_APP, EMAIL

3.2 事件處理流程



4. 資料庫設計

```
-- 通知範本表
CREATE TABLE notification_templates (
    template_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    template_code VARCHAR(100) NOT NULL UNIQUE,
    template_name VARCHAR(255) NOT NULL,
    subject VARCHAR(500),
    body TEXT NOT NULL,
    default_channels JSONB DEFAULT '["IN_APP"]',
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 通知記錄表
CREATE TABLE notifications (
    notification_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    recipient_id UUID NOT NULL,
    title VARCHAR(500) NOT NULL,
    content TEXT NOT NULL,
    notification_type VARCHAR(30) NOT NULL
        CHECK (notification_type IN ('APPROVAL_REQUEST', 'APPROVAL_RESULT', 'REMINDER', 'AN
    channels JSONB DEFAULT '["IN_APP"]',
    priority VARCHAR(20) DEFAULT 'NORMAL' CHECK (priority IN ('LOW', 'NORMAL', 'HIGH', 'URG
    status VARCHAR(20) DEFAULT 'PENDING' CHECK (status IN ('PENDING', 'SENT', 'FAILED', 'RE
    sent_at TIMESTAMP,
    read_at TIMESTAMP,
    related_business_type VARCHAR(50),
    related_business_id UUID,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_notification_recipient ON notifications(recipient_id, status);
CREATE INDEX idx_notification_created ON notifications(created_at DESC);

-- 通知偏好設定表
CREATE TABLE notification_preferences (
    preference_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    employee_id UUID NOT NULL UNIQUE,
    email_enabled BOOLEAN DEFAULT TRUE,
    push_enabled BOOLEAN DEFAULT TRUE,
    in_app_enabled BOOLEAN DEFAULT TRUE,
    quiet_hours_start TIME,
    quiet_hours_end TIME,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

5. 自動提醒Job

Job名稱	執行頻率	觸發事件
`BirthdayReminderJob`	每日08:00	發送當日生日祝福
`ContractExpiryJob`	每日09:00	合約30天內到期提醒HR
`CertificateExpiryJob`	每週一	證照30天內到期提醒員工
`AnnualLeaveExpiryJob`	每週一	特休30天內到期提醒
`TimesheetReminderJob`	每日18:00	未回報工時提醒

6. API設計 (8個端點)

端點	方法	Controller
`/api/v1/notifications/send`	POST	HR12NotificationCmdController
`/api/v1/notifications/me`	GET	HR12NotificationQryController
`/api/v1/notifications/{id}/read`	PUT	HR12NotificationCmdController
`/api/v1/notifications/read-all`	PUT	HR12NotificationCmdController
`/api/v1/notifications/unread-count`	GET	HR12NotificationQryController
`/api/v1/notifications/templates`	POST	HR12TemplateCmdController
`/api/v1/notifications/templates`	GET	HR12TemplateQryController
`/api/v1/notifications/preferences`	PUT	HR12PreferenceCmdController

7. Domain設計

7.1 Notification聚合根 (通知)

職責: 管理通知的建立、發送與狀態追蹤

Java實作:

```java
@Entity
@Table(name = "notifications")
public class Notification {
    @EmbeddedId
    private NotificationId id;

    @Column(name = "recipient_id", nullable = false)
    private UUID recipientId;

    @Column(name = "title", nullable = false, length = 500)
    private String title;

    @Column(name = "content", nullable = false, columnDefinition = "TEXT")
    private String content;

    @Enumerated(EnumType.STRING)
    @Column(name = "notification_type", nullable = false)
    private NotificationType notificationType;

    @Convert(converter = ChannelListConverter.class)
    @Column(name = "channels", columnDefinition = "JSONB")
    private List<NotificationChannel> channels;

    @Enumerated(EnumType.STRING)
    @Column(name = "priority", nullable = false)
    private Priority priority;

    @Enumerated(EnumType.STRING)
    @Column(name = "status", nullable = false)
    private NotificationStatus status;

    @Column(name = "sent_at")
    private LocalDateTime sentAt;

    @Column(name = "read_at")
    private LocalDateTime readAt;

    @Column(name = "related_business_type", length = 50)
    private String relatedBusinessType;

    @Column(name = "related_business_id")
    private UUID relatedBusinessId;

    // ========== Factory Method ==========

    /**
     * 建立通知
     */
    public static Notification create(
            UUID recipientId,
            String title,
```

```
            String content,
            NotificationType type,
            List<NotificationChannel> channels,
            Priority priority) {

        Notification notification = new Notification();
        notification.id = NotificationId.generate();
        notification.recipientId = recipientId;
        notification.title = Objects.requireNonNull(title, "標題不可為空");
        notification.content = Objects.requireNonNull(content, "內容不可為空");
        notification.notificationType = type;
        notification.channels = channels != null ? channels : List.of(NotificationChannel.I
        notification.priority = priority != null ? priority : Priority.NORMAL;
        notification.status = NotificationStatus.PENDING;

        // 發布通知建立事件
        DomainEventPublisher.publish(new NotificationCreatedEvent(
            notification.id.getValue(),
            notification.recipientId,
            notification.channels
        ));

        return notification;
    }

    /**
     * 標記為已發送
     */
    public void markAsSent() {
        if (this.status != NotificationStatus.PENDING) {
            throw new DomainException("只有待發送的通知可以標記為已發送");
        }

        this.status = NotificationStatus.SENT;
        this.sentAt = LocalDateTime.now();
    }

    /**
     * 標記發送失敗
     */
    public void markAsFailed(String errorMessage) {
        this.status = NotificationStatus.FAILED;
        // 記錄錯誤日誌
        DomainEventPublisher.publish(new NotificationFailedEvent(
            this.id.getValue(),
            this.recipientId,
            errorMessage
        ));
    }

    /**
```

```java
     * 標記為已讀
     */
    public void markAsRead() {
        if (this.status != NotificationStatus.SENT) {
            throw new DomainException("只有已發送的通知可以標記為已讀");
        }

        this.status = NotificationStatus.READ;
        this.readAt = LocalDateTime.now();

        DomainEventPublisher.publish(new NotificationReadEvent(
            this.id.getValue(),
            this.recipientId
        ));
    }

    /**
     * 設定業務關聯
     */
    public void setBusinessRelation(String businessType, UUID businessId) {
        this.relatedBusinessType = businessType;
        this.relatedBusinessId = businessId;
    }

    /**
     * 檢查是否未讀
     */
    public boolean isUnread() {
        return this.status == NotificationStatus.SENT;
    }
}

// 值對象
public enum NotificationType {
    APPROVAL_REQUEST,    // 審核請求
    APPROVAL_RESULT,     // 審核結果
    REMINDER,            // 提醒
    ANNOUNCEMENT,        // 公告
    ALERT                // 警示
}

public enum NotificationChannel {
    IN_APP,   // 系統內通知
    EMAIL,    // Email
    PUSH,     // 推播
    TEAMS,    // Microsoft Teams
    LINE      // LINE
}

public enum Priority {
    LOW,
```

```
    NORMAL,
    HIGH,
    URGENT
}

public enum NotificationStatus {
    PENDING,   // 待發送
    SENT,      // 已發送
    FAILED,    // 發送失敗
    READ       // 已讀
}
```

7.2 NotificationTemplate實體 (通知範本)

```java
@Entity
@Table(name = "notification_templates")
public class NotificationTemplate {
    @EmbeddedId
    private TemplateId id;

    @Column(name = "template_code", nullable = false, unique = true, length = 100)
    private String templateCode;

    @Column(name = "template_name", nullable = false, length = 255)
    private String templateName;

    @Column(name = "subject", length = 500)
    private String subject;

    @Column(name = "body", nullable = false, columnDefinition = "TEXT")
    private String body;

    @Convert(converter = ChannelListConverter.class)
    @Column(name = "default_channels", columnDefinition = "JSONB")
    private List<NotificationChannel> defaultChannels;

    @Column(name = "is_active")
    private boolean isActive;

    /**
     * 替換變數
     */
    public String renderContent(Map<String, Object> variables) {
        String rendered = this.body;

        for (Map.Entry<String, Object> entry : variables.entrySet()) {
            String placeholder = "{{" + entry.getKey() + "}}";
            rendered = rendered.replace(placeholder, String.valueOf(entry.getValue()));
        }

        return rendered;
    }

    /**
     * 替換主旨變數
     */
    public String renderSubject(Map<String, Object> variables) {
        if (this.subject == null) return null;

        String rendered = this.subject;
        for (Map.Entry<String, Object> entry : variables.entrySet()) {
            String placeholder = "{{" + entry.getKey() + "}}";
            rendered = rendered.replace(placeholder, String.valueOf(entry.getValue()));
```

```
        }

        return rendered;
    }
}
```

7.3 NotificationPreference實體 (通知偏好)

```
@Entity
@Table(name = "notification_preferences")
public class NotificationPreference {
    @EmbeddedId
    private PreferenceId id;

    @Column(name = "employee_id", nullable = false, unique = true)
    private UUID employeeId;

    @Column(name = "email_enabled")
    private boolean emailEnabled;

    @Column(name = "push_enabled")
    private boolean pushEnabled;

    @Column(name = "in_app_enabled")
    private boolean inAppEnabled;

    @Column(name = "quiet_hours_start")
    private LocalTime quietHoursStart;

    @Column(name = "quiet_hours_end")
    private LocalTime quietHoursEnd;

    /**
     * 過濾通知渠道
     */
    public List<NotificationChannel> filterChannels(List<NotificationChannel> requestedChan
        List<NotificationChannel> filtered = new ArrayList<>();

        for (NotificationChannel channel : requestedChannels) {
            if (isChannelEnabled(channel)) {
                filtered.add(channel);
            }
        }

        // 至少保留系統內通知
        if (filtered.isEmpty()) {
            filtered.add(NotificationChannel.IN_APP);
        }

        return filtered;
    }

    private boolean isChannelEnabled(NotificationChannel channel) {
        return switch (channel) {
            case EMAIL -> emailEnabled;
            case PUSH -> pushEnabled;
            case IN_APP -> inAppEnabled;
```

```
            default -> true; // TEAMS, LINE 預設開啟
        };
    }

    /**
     * 檢查是否在靜音時段
     */
    public boolean isInQuietHours() {
        if (quietHoursStart == null || quietHoursEnd == null) {
            return false;
        }

        LocalTime now = LocalTime.now();
        return !now.isBefore(quietHoursStart) && !now.isAfter(quietHoursEnd);
    }
}
```

---

8. 領域事件設計

8.1 事件總覽

事件名稱	觸發時機	訂閱服務	重要性
`NotificationCreatedEvent`	通知建立	Notification內部	⭐⭐
`NotificationSentEvent`	通知發送成功	Notification內部	⭐⭐
`NotificationFailedEvent`	通知發送失敗	Monitoring	⭐⭐⭐
`NotificationReadEvent`	通知已讀	Notification內部	⭐

注意: Notification Service主要作為**事件訂閱者**,訂閱其他業務服務的事件,較少發布自己的領域事件。

8.2 訂閱的業務事件

Notification Service訂閱以下業務事件並發送通知:

業務事件	來源服務	通知對象	範本代碼
`LeaveAppliedEvent`	Attendance	直屬主管	LEAVE_APPROVAL_REQUEST
`LeaveApprovedEvent`	Attendance	申請人	LEAVE_APPROVED
`LeaveRejectedEvent`	Attendance	申請人	LEAVE_REJECTED
`PayslipGeneratedEvent`	Payroll	員工	PAYSLIP_READY
`TaskAssignedEvent`	Workflow	審核人	APPROVAL_REQUEST
`EmployeeCreatedEvent`	Organization	新員工	WELCOME_MESSAGE
`ContractExpiringEvent`	Organization	HR	CONTRACT_EXPIRY
`CertificateExpiringEvent`	Training	員工	CERTIFICATE_EXPIRY

8.3 事件Schema

8.3.1 NotificationCreatedEvent

```json
{
  "eventId": "evt-ntf-001",
  "eventType": "NotificationCreated",
  "timestamp": "2025-12-06T10:00:00Z",
  "payload": {
    "notificationId": "550e8400-e29b-41d4-a716-446655440000",
    "recipientId": "emp-001",
    "channels": ["IN_APP", "EMAIL"],
    "priority": "NORMAL"
  }
}
```

8.3.2 NotificationFailedEvent (關鍵事件)

```
{
  "eventId": "evt-ntf-002",
  "eventType": "NotificationFailed",
  "timestamp": "2025-12-06T10:05:00Z",
  "payload": {
    "notificationId": "550e8400-e29b-41d4-a716-446655440001",
    "recipientId": "emp-002",
    "channel": "EMAIL",
    "errorCode": "SMTP_CONNECTION_FAILED",
    "errorMessage": "SMTP伺服器連線逾時",
    "retryCount": 3,
    "willRetry": false
  }
}
```

訂閱者處理:

- **Monitoring Service:** 記錄錯誤日誌、發送告警

---

9. 事件訂閱實作範例

9.1 訂閱LeaveApprovedEvent發送通知

```
@Service
@Slf4j
public class LeaveEventListener {
    @Autowired
    private INotificationTemplateRepository templateRepository;

    @Autowired
    private NotificationSender notificationSender;

    @Autowired
    private INotificationPreferenceRepository preferenceRepository;

    @KafkaListener(topics = "leave.approved")
    public void handleLeaveApproved(LeaveApprovedEvent event) {
        try {
            // 1. 載入通知範本
            NotificationTemplate template = templateRepository
                .findByTemplateCode("LEAVE_APPROVED");

            // 2. 準備變數
            Map<String, Object> variables = Map.of(
                "employeeName", event.getEmployeeName(),
                "leaveType", event.getLeaveTypeName(),
                "startDate", event.getStartDate(),
                "endDate", event.getEndDate(),
                "totalDays", event.getTotalDays()
            );

            // 3. 渲染內容
            String title = "請假申請已核准";
            String content = template.renderContent(variables);

            // 4. 檢查用戶偏好
            NotificationPreference preference = preferenceRepository
                .findByEmployeeId(event.getEmployeeId());

            List<NotificationChannel> channels = preference.filterChannels(
                List.of(NotificationChannel.IN_APP, NotificationChannel.EMAIL)
            );

            // 5. 建立通知
            Notification notification = Notification.create(
                event.getEmployeeId(),
                title,
                content,
                NotificationType.APPROVAL_RESULT,
```

```
                channels,
                Priority.NORMAL
            );

            notification.setBusinessRelation("LEAVE_APPLICATION", event.getApplicationId())

            notificationRepository.save(notification);

            // 6. 發送通知
            notificationSender.send(notification);

            log.info("Successfully sent leave approved notification to employee: {}",
                event.getEmployeeId());

        } catch (Exception e) {
            log.error("Failed to send leave approved notification", e);
        }
    }
}
```

9.2 多渠道發送Service

```java
@Service
public class NotificationSender {
    @Autowired
    private EmailSender emailSender;

    @Autowired
    private WebSocketSender webSocketSender;

    @Autowired
    private PushNotificationSender pushSender;

    @Autowired
    private INotificationRepository notificationRepository;

    public void send(Notification notification) {
        boolean allSuccess = true;

        for (NotificationChannel channel : notification.getChannels()) {
            try {
                switch (channel) {
                    case IN_APP -> sendInApp(notification);
                    case EMAIL -> sendEmail(notification);
                    case PUSH -> sendPush(notification);
                    case TEAMS -> sendTeams(notification);
                    case LINE -> sendLine(notification);
                }
            } catch (Exception e) {
                log.error("Failed to send via {}: {}", channel, e.getMessage());
                allSuccess = false;
            }
        }

        if (allSuccess) {
            notification.markAsSent();
        } else {
            notification.markAsFailed("部分渠道發送失敗");
        }

        notificationRepository.save(notification);
    }

    private void sendInApp(Notification notification) {
        // WebSocket 即時推送
        webSocketSender.sendToUser(
            notification.getRecipientId(),
            notification
        );
    }
```

```
    private void sendEmail(Notification notification) {
        emailSender.send(
            notification.getRecipientId(),
            notification.getTitle(),
            notification.getContent()
        );
    }

    private void sendPush(Notification notification) {
        pushSender.send(
            notification.getRecipientId(),
            notification.getTitle(),
            notification.getContent()
        );
    }
}
```

10. 工項清單摘要

10.1 前端開發工項

工項編號	工項名稱	說明	預估工時 (人天)
FE-NTF-01	HR12-P01 通知範本管理頁面	範本CRUD	2
FE-NTF-02	HR12-P02 我的通知頁面	通知列表、已讀/未讀	3
FE-NTF-03	HR12-P03 通知偏好設定頁面	偏好設定CRUD	2
FE-NTF-04	通知中心Widget	頂部通知鈴鐺、下拉列表	2
FE-NTF-05	WebSocket整合	即時通知推送	2
FE-NTF-06	Redux狀態管理	Notification State、Actions	1
FE-NTF-07	API Service層	NotificationService、API封裝	1
FE-NTF-08	Factory層	ViewModel轉換	1
FE-NTF-09	單元測試	Factory、Component測試	1
小計			**15人天**

10.2 後端開發工項

工項編號	工項名稱	說明	預估工時 (人天)
BE-NTF-01	Notification聚合根	通知Domain邏輯	2
BE-NTF-02	NotificationTemplate實體	範本管理、變數替換	2
BE-NTF-03	NotificationPreference實體	偏好設定邏輯	1
BE-NTF-04	Repository層	3個Repository	1
BE-NTF-05	通知API	查詢/已讀API (5端點)	2
BE-NTF-06	範本管理API	範本CRUD (2端點)	1
BE-NTF-07	偏好設定API	偏好CRUD (1端點)	1
BE-NTF-08	多渠道發送Service	Email/Push/Teams/LINE整合	5
BE-NTF-09	事件訂閱Listener	訂閱8個業務事件	4
BE-NTF-10	WebSocket支援	STOMP協定即時推送	3
BE-NTF-11	自動提醒Job	5個定時任務 (生日、合約到期等)	3
BE-NTF-12	Email範本引擎	Thymeleaf HTML範本	2
BE-NTF-13	單元測試	Domain層100%覆蓋率	2
BE-NTF-14	整合測試	API端點測試	2
小計			**31人天**

10.3 第三方整合工項

工項編號	工項名稱	說明	預估工時 (人天)
INT-NTF-01	SMTP Email整合	Spring Mail設定	1
INT-NTF-02	Firebase FCM整合	iOS/Android推播	3
INT-NTF-03	Microsoft Teams Webhook	Teams通知	2
INT-NTF-04	LINE Notify API	LINE通知	2
小計			**8人天**

10.4 測試工項

工項編號	工項名稱	說明	預估工時 (人天)
TEST-NTF-01	Domain單元測試	Notification、Template、Preference	2
TEST-NTF-02	API整合測試	8個端點	1
TEST-NTF-03	事件訂閱測試	8個Listener	2
TEST-NTF-04	多渠道發送測試	Mock第三方服務	2
TEST-NTF-05	E2E測試	完整通知流程	2
小計			**9人天**

10.5 文件工項

工項編號	工項名稱	預估工時 (人天)
DOC-NTF-01	API文件 (Swagger)	1
DOC-NTF-02	通知範本開發指南	1
DOC-NTF-03	第三方整合文件	1
小計		**3人天**

10.6 總工時估計

類別	工時 (人天)
前端開發	15
後端開發	31
第三方整合	8
測試	9
文件	3
總計	**66人天**

備註:

- Email範本建議使用Thymeleaf,支援HTML格式與變數替換
- WebSocket連線管理需考慮斷線重連機制
- 第三方服務整合建議實作重試機制與降級策略

- 建議分兩階段實作:
  - **階段一:** 基礎通知 (系統內通知 + Email) - 35人天
  - **階段二:** 進階功能 (Push + Teams + LINE + 自動提醒) - 31人天

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